SUNIL KHATRI

Ajmer, Rajasthan, India | +91 9672983175 | sunilkhatri15@gmail.com | linkedin.com/in/sunilk15 | sunilkhatri.wiki | github.com/Sunil1234267

Full-Stack Software Engineer • Embedded Systems & Hardware Architect • Generative AI / ML • Automotive Electronics (BMS / VCU / BCM)

PROFESSIONAL SUMMARY

Depth-first Systems Engineer with 4+ years of hands-on production engineering experience spanning automotive embedded platforms (2021–2025 at **Pravaig Dynamics** and **KPIT Technologies**) and modern cloud-native / Generative AI systems (2023–Present as an autonomous founder and systems architect). Proven track record in safety-critical vehicle electrification: led full ASW development for EV BMS, VCU, and BCM using Extended Kalman Filters for LFP chemistries at Pravaig, and led Mercedes-Benz SIL/MIL testing plus TVS Motors Renesas/ESP32 sensor firmware development at KPIT. Independently designed, architected, and shipped four complete production platforms: **VanadiumVortex** (Kubernetes-native AI fabric e-commerce platform with KEDA scale-to-zero GPU autoscaling and 10+ services), **MotorCyclePro** (custom multi-layer AEC-Q100 automotive PCB, ESP32-S3 firmware, sub-65 µA deep sleep, LoRa/CAN/BLE, and Android/WearOS app), **STP-AI** (FLUX.2 Klein 4B generative pipeline delivering 600 DPI rotary-screen-print-ready textile CAD), and **STP Receipt System** (industrial IoT print management system deployed on Raspberry Pi for an active manufacturing plant). Distinguished by first-principles root-cause engineering, adversarial threat modeling, rigorous MISRA-C and automotive compliance, and quantitative decision-making measured to µA, millivolt, and cent precision.

TECHNICAL SKILLS MATRIX

Languages	C, C++ (Embedded, MISRA-C:2012), Python 3, JavaScript / TypeScript, Kotlin, Bash, SQL
Embedded & MCU	ESP32-S3 / C3, STM32, Ti TMS540, Renesas RH850; FreeRTOS, ESP-IDF; BLE 5.0, LoRa SX1262 (IN865), CAN 2.0B / TWAI, LIN, I2C, SPI, UART, ADC, PWM, Dual-slot OTA rollback; Ultra-low-power design (sub-65 µA baseline)
Automotive Systems	BMS (SoC / SoH / SoE, Extended Kalman Filter), VCU, BCM; CAN/LIN network management, UDS diagnostics (ISO 14229), DBC authoring; AUTOSAR methodology; Vector CANoe, CANalyzer, PikeTec TPT (SIL / MIL); MATLAB & Simulink
Hardware & PCB	KiCad (schematic capture, 4-layer impedance-matched layout); AEC-Q100 / Q200 automotive component selection; High-side smart switches (PROFET BTS7008), ISO 7637-2 load-dump TVS protection, 50Ω coplanar RF waveguides
Cloud & DevOps	Kubernetes (k3s), Docker, Docker Compose; KEDA (HTTP & Redis queue autoscaling, scale-to-zero GPU pods); NGINX Ingress, cert-manager (automated Let's Encrypt TLS); Kustomize, GitHub Actions CI/CD
Backend & Distributed	Node.js, Express.js; PostgreSQL 16 (pg-boss background jobs, LISTEN / NOTIFY event bus, UUID PKs, advisory locks); Redis 7; MinIO S3-compatible storage (100 Gi PVC); Microservices monorepo; REST, SSE, WebSockets
Frontend & Mobile	React 18, TypeScript, Vite, Zustand, Radix UI, Tailwind CSS; Android (Kotlin, Jetpack Compose, Material3, 120 fps RenderNode drag engine, FLIP animations, SAF Drive backup); Wear OS; Android Auto (DHU)
Generative AI & ML	PyTorch, Hugging Face Diffusers; FLUX.2 Klein 4B; DreamBooth fine-tuning; Multi-LoRA dynamic adapter blending; 4-bit NF4 quantization (bitsandbytes, device_map=auto); Swin2SR 4x super-resolution; Modal cloud GPU (B200 / H100)
Security & Cryptography	AES-GCM (ESP32-S3 hardware cryptographic accelerator), HMAC-SHA256, SuperTokens Core, JWT; Rolling NVS nonces (replay protection), Challenge-response pairing, Rate limiting, SSRF defense, Signed imgproxy URLs
IoT & Systems	Linux systemd service architecture, Raspberry Pi 4, ESC/POS thermal printing protocol, BLE daemon, Winston structured JSON logging, Local LLM deployment (Llama 3.1 8B on RTX 4060 with Docker & NF4)

PROFESSIONAL EXPERIENCE

Senior Embedded Software Engineer — KPIT Technologies **March 2024 – August 2025**

Automotive Powertrain & Sensor Systems | Clients: Mercedes-Benz R&D • TVS Motors Ltd. *Pune & Bangalore, India*

- **Mercedes-Benz R&D (SIL / MIL Test Campaigns):** Orchestrated Software-in-the-Loop and Model-in-the-Loop testing campaigns for automotive powertrain ECUs using PikeTec TPT; authored regression test suites improving fault-injection test coverage by 30% and accelerating integration sign-off cycles.
- **TVS Motors (Vehicle Sensor Platform Bring-Up):** Led low-level firmware bring-up and peripheral driver engineering for next-generation vehicle sensor suites across Renesas RH850 and ESP32 platforms; took architecture from register-level bare-metal bring-up through RTOS task scheduling to production release.
- **IMU Lean Angle Sensor Innovation:** Designed and deployed a proprietary 6-axis IMU sensor fusion algorithm for motorcycle lean detection; modeled sensor dynamics in MATLAB/Simulink and deployed optimized C firmware to ESP32 with sub-millisecond interrupt response, achieving marked noise reduction over mechanical switch alternatives.
- **Automotive Quality & Compliance:** Enforced strict **MISRA-C:2012 compliance** across all delivered firmware modules; executed cross-functional hardware-in-the-loop (HIL) validation using Vector CANoe, CANalyzer, mixed-signal oscilloscopes, and logic analyzers.

Electronics & Embedded Engineer — Pravaig Dynamics Pvt. Ltd. **December 2021 – December 2023**

Electric Vehicle (EV) Platform Division | Core Powertrain & Body Electronics *Bengaluru, India*

- **Full ASW Ownership (BMS / VCU / BCM):** Owned the complete application-software (ASW) stack for the Battery Management System, Vehicle Control Unit, and Body Control Module; drove vehicle electrification from bare PCB bring-up to customer-ready roadworthy prototypes.
- **Advanced BMS State Estimation:** Architected and coded production BMS algorithms — **State of Charge (SoC), State of Health (SoH), and State of Energy (SoE)** — utilizing **Extended Kalman Filtering (EKF)** tailored to non-linear LFP cell discharge curves; delivered active cell-balancing routines, pre-charge sequencing, and multi-tier thermal runaway protection.
- **Vehicle Networking & Diagnostics:** Developed complete CAN 2.0B / LIN communication drivers and **UDS diagnostic services (ISO 14229)** for cross-ECU telemetry; created vehicle-wide DBC database files and conducted end-to-end bus load analysis in Vector CANoe.
- **Hardware-Software Integration:** Collaborated daily with hardware design and mechanical packaging teams to guide schematic revisions, thermal layout, and connector selection; resolved high-voltage grounding loops and EMI transients under aggressive track testing.

Founder & Principal Systems Engineer — Independent Projects **2023 – Present**

Full-Stack Web • Generative AI • Embedded Hardware • Cloud Infrastructure *Ajmer & Remote, India*

- **Autonomous System Lifecycle:** Conceived, designed, built, and shipped four enterprise-grade production systems solo — spanning automotive-grade PCB fabrication, embedded firmware, mobile companion apps, generative AI inference pipelines, and distributed Kubernetes cloud clusters.
- **Forensic Database & Security Auditing:** Conducted exhaustive multidimensional security and concurrency audits across own production systems; remediated **111 forensic database defects** covering UUID migration, pg_advisory_lock distributed race-condition prevention, and connection-pool serialization.

EDUCATION

B.Tech — Electrical & Electronics Engineering **2017 – 2021**

SRM Institute of Science and Technology, Chennai, India *Chennai, India*

CERTIFICATIONS & SPECIALIZED TRAINING

• **EV Powertrain & Embedded Systems Specialization**	Autobot Academy (India's Premier EV Engineering Bootcamp)	**April 2021**
• **CAN Bus Protocol & In-Vehicle Networking**	Professional Automotive Engineering Certification	**2022**
• **IoT & Embedded Systems Specialization**	University of California, Irvine (Coursera)	**2021**
• **Python for Everybody Specialization**	University of Michigan (Coursera)	**2021**
• **Learn Industrial Automation**	LinkedIn Learning	**September 2020**
• **Learning FPGA Development**	LinkedIn Learning	**October 2020**
• **MATLAB 2018 Essential Training**	LinkedIn Learning	**October 2020**

FEATURED PRODUCTION PROJECTS • FLAGSHIP SYSTEMS

VanadiumVortex — AI-Powered Fabric E-Commerce & ERP Platform

March 2025 – Present

Production Cloud Deployment: vanadiumvortex.com | Full-Stack • Microservices • Cloud • GenAI

Milestone: Forensic Overhaul Sept 2026

Node.js • PostgreSQL 16 • Redis 7 • Kubernetes (k3s) • KEDA • Python GPU Engine • MinIO • SuperTokens • imgproxy • Docker

- **Kubernetes-Native Infrastructure:** Engineered and deployed a 29-resource production **k3s cluster** featuring NGINX Ingress, automated Let's Encrypt TLS (cert-manager), and zero-downtime rolling deployments; configured **KEDA scale-to-zero GPU workers** triggered by Redis job queue depth — reducing idle GPU hosting costs to $0.
- **Microservices Event Architecture:** Built 5 decoupled Node.js services (Catalog, Order, Auth, AI Chatbot, Razorpay Payment Bridge) communicating asynchronously over **PostgreSQL LISTEN/NOTIFY** — eliminating HTTP coupling, webhook fragility, and intermediary message broker overhead.
- **Dual-Mode Auth & Enterprise Security:** Deployed SuperTokens Core alongside backward-compatible legacy JWT authentication with automated bcrypt password re-hashing hooks; established granular RBAC (Admin, Staff, Finance, Logistics), IP-rate-limiting, SSRF defense, and **HMAC-SHA256 signed imgproxy URLs** to prevent Denial-of-Wallet image processing attacks.
- **High-Capacity Storage & Logistics:** Integrated self-hosted MinIO S3 object storage (100 Gi PVC across 5 dedicated buckets) with on-the-fly WebP/AVIF transformation; implemented multi-courier fulfillment (Shiprocket / Easyship) through persistent pg-boss background workers with automated currency conversion.
- **111-Defect Forensic Overhaul:** Executed a formal multidimensional audit resolving 111 database issues: UUID primary keys against enumeration attacks, distributed advisory locking on migrations, named SQL parameter translation, and SQLite serialization guarantees on pooled PostgreSQL.

MotorCyclePro — Automotive Smart Controller & Mobile Ecosystem

2023 – Present

Hardware Iterations V1 (2023) • V2 (2024) • V3 Production PCB & Companion App (2026)

Scorecard: 130/130 Automotive Audited

KiCad • ESP32-S3-MINI-1U-N8 • BLE 5.0 • LoRa SX1262 (IN865) • CAN 2.0B • ASM330LHHTR IMU • Quectel L89HA • Kotlin (Android / Wear OS)

- **Automotive PCB Engineering (KiCad V1–V3):** Designed a custom multi-layer automotive controller PCB; integrated **ESP32-S3-MINI-1U-N8** (dual-core LX7 @ 240 MHz), **SX1262 LoRa** (865–867 MHz IN865 band), **ASM330LHHTR 6-axis automotive IMU** (with Machine Learning Core for autonomous tip detection), **Quectel L89HA GNSS/IRNSS**, and **TCAN1044A-Q1 CAN transceiver** with AEC-Q100/Q200 components and conformal coating.
- **Hardware Review & Failure Mode Rectification:** Authored an authoritative hardware audit resolving 11 critical design vulnerabilities: eliminated ESP32-S3 GPIO over-subscription (reconciled 33 GPIOs to 28 core + 4 add-ons); corrected load-dump TVS protection (replaced SMCJ24A clamping at 38.9 V with SMCJ20A to stay strictly under buck regulator LMR33630's 36 V abs-max); re-engineered cranking UVLO with 0.5 V hysteresis to prevent reboot loops during 8V engine cranking; corrected LiFePO4 charging chemistry.
- **Ultra-Low-Power Sleep (<65 µA):** Engineered an ultra-low-power quiescent current budget drawing only ~65 µA in deep sleep (~1.5% battery capacity / month on a 10 Ah motorcycle battery); implemented multi-source RTC wake (V_IGN ADC, IMU INT1, LoRa DIO0). Replaced continuous LoRa CAD with hardware SetRxDutyCycle, preventing the 260 mAh/day drain of un-gated listening.
- **Cryptographic Wire Protocol:** Built an AES-128-GCM authenticated BLE communication protocol using the ESP32-S3 hardware crypto accelerator; implemented rolling NVS nonces to defeat replay attacks and HMAC challenge-response pairing to prevent relay theft. Formulated a 25-command binary wire protocol with START_BYTE 0xAA, 16-bit payload length, and CRC8-CCITT.
- **Android, Wear OS & Auto Companion Suite:** Created a high-performance companion app in Kotlin / Jetpack Compose / Material3; engineered a 120 fps hardware-RenderNode drag engine with FLIP spring physics, Storage Access Framework (SAF) JSON cloud backups, live telemetry gauges, Android Auto (DHU) projection, and dual-slot OTA firmware updates with CRC32 verification and automatic rollback.

FEATURED PRODUCTION PROJECTS • AI & INDUSTRIAL IOT

STP-AI — Industrial Textile CAD Generative AI Engine

March 2025 – Present

Production Inference Studio • 6-Script DreamBooth Pipeline | Sunil Textile Print *B200 Cloud GPU Integration (2026)*

Python • PyTorch • FLUX.2 Klein 4B • DreamBooth • LoRA • Swin2SR 4x • bitsandbytes NF4 • Flask • Modal Cloud GPU • Hugging Face

- **Quantized Production Diffusion:** Tailored **FLUX.2 Klein 4B** using **4-bit NF4 quantization** (bitsandbytes, device_map=auto) for single-GPU production serving; built a thread-safe Singleton model manager with double-checked locking to prevent VRAM allocation collisions.
- **Dynamic Multi-LoRA Adapter Blending:** Architected a dynamic LoRA injection layer allowing real-time stylistic cross-blending across textile domains (floral, mandala, damask baroque, geometric) without incurring model reload latency.
- **Industrial Print CAD Processing (600 DPI):** Engineered an end-to-end Python image synthesis pipeline tailored for rotary screen printing: bilateral vector line sharpening, continuous raised-cosine seam blending for infinite roll repeat, 4-way bookmatch tiling, precision spot-color indexing with area coverage using ITU-R BT.709 perceptual luminance (0.2126 R + 0.7152 G + 0.0722 B), and dual-format 600 DPI export (lossless PNG + TIFF/LZW).
- **Super-Resolution & Automated Training:** Integrated **Swin2SR 4x super-resolution** for ultra-crisp pattern scaling; developed a 6-script automated DreamBooth training pipeline handling dataset hygiene, synthetic captioning, auto-sorting, Modal cloud GPU launches (targeting NVIDIA B200 / H100), and Hugging Face model checkpoint publishing.

STP Receipt System — Industrial IoT Print Management Platform

September 2025 – Present

On-Premises Production Deployment on Raspberry Pi (stp.local) | Factory Operations *Continuous Production (2025–2026)*

React 18 • TypeScript • Vite • Zustand • Radix UI • Node.js • Express • Winston • ESC/POS • Raspberry Pi • BLE • systemd

- **Industrial On-Premises Architecture:** Designed and deployed a full-stack factory operations and thermal receipt platform on a Raspberry Pi 4 operating 24/7 on the local manufacturing intranet at stp.local.
- **Zero-Configuration Printer Discovery:** Implemented automated LAN ESC/POS network printer discovery via subnet IP sweep and /health probe verification; integrated Server-Sent Events (SSE) for zero-downtime client telemetry, alongside a background Bluetooth BLE daemon managed as a Linux systemd unit.
- **Fault-Tolerant Reliability Layer:** Production-hardened with Winston structured JSON logging, daily log rotation, automatic error classification (Validation / Network / System / Operation), and error-spike alerts (>10 errors/min). Built typed error handling (ApiError, TimeoutError), 3-attempt exponential backoff retry, React error boundaries, and offline log queuing.

ADDITIONAL TECHNICAL PROJECTS & RESEARCH

Business Process Automation (n8n)

2024

- Architected scalable n8n workflow automations bridging enterprise finance, sales CRM, and textile warehouse inventory databases; deployed self-hosted instances on Linux with automated daily backup routines.

Local LLM Inference Engine (Llama 3.1 8B)

2024

- Quantized and deployed Llama 3.1 8B on local hardware (NVIDIA RTX 4060) utilizing Docker, CUDA acceleration, and 4-bit NF4 quantization; tuned context caching and generation throughput for low-latency assistant queries.

IMU Sensor Fusion System (Vehicle Dynamics)

2024

- Developed an ESP32-based Kalman-filtered lean angle estimation system with sub-millisecond interrupt response; calibrated complementary filtering to reject high-frequency road vibrations while tracking dynamic lean transitions.

Automotive Multi-Layer PCB (KiCad BMW Platform)

2025 – 2026

- Engineered an automotive-grade daughterboard in KiCad featuring TE Superseal / Deutsch DT connector footprints, ground plane isolation, ISO 7637-2 pulse suppression, and high-current copper thermal vias.

ENGINEERING MINDSET & SYSTEMS APPROACH

- **First-Principles, Root-Cause Engineering:** Resolves complex problems at physical, electrical, and logical root levels rather than applying surface patches. Example: Proved that a standard SMCJ24A TVS diode fails to protect an LMR33630 buck regulator during automotive load dump because its 38.9 V clamping voltage exceeds the IC's 36 V absolute-maximum rating — mathematically deriving the requirement for an SMCJ20A replacement.
- **Cross-Domain Systems Synthesis:** Seamlessly reasons across mechanical vibration, silicon physics, low-level firmware, distributed clouds, and user interaction. Example: Computed that motorcycle IMU rubber isolation resonance (10–30 Hz) must sit strictly above engine idle but below vehicle tip-over frequency to prevent Machine Learning Core (MLC) false positives.
- **Adversarial & Security-First Threat Modeling:** Analyzes multi-vector vulnerabilities at architectural conception — identifying that proximity BLE beacons invite relay attacks (replaced with HMAC challenge-response), that RTC deep-sleep clock drift causes timestamp replay checks to fail open (replaced with NVS rolling nonces), and that unprotected image transcoding creates Denial-of-Wallet attack surfaces.
- **Quantitative Precision & Standards Compliance:** Every design choice is anchored in empirical calculations: quiescent sleep budgets to µA precision (65 µA baseline vs. 5 mA continuous CAN drain), GPU training cost tradeoffs (T4 vs B200), and UVLO cranking hysteresis. Strictly adheres to formal standards (ITU-R BT.709, AEC-Q100, ISO 14229, ISO 7637-2, MISRA-C:2012) over informal approximations.